

Q2A: Derive analytically the property of Fourier transform for the derivative of a function. Apply this property to find the Fourier transform of a specific function and its derivative, providing numerical calculations to support the result.

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Part 1: Derivation of the Derivative Property

Definition

The Fourier transform of a function $f(x)$ is defined as

$$\hat{f}(k) = \mathcal{F}[f](k) = \int_{-\infty}^{\infty} f(x) e^{ikx} dx.$$

Theorem 1 (Derivative Property). *Let f be a continuously differentiable, absolutely integrable function on $\mathbb{R}$ with $\lim_{x \rightarrow \pm\infty} f(x) = 0$. Then*

$$\mathcal{F}[f'](k) = -ik \hat{f}(k).$$

Proof. By definition,

$$\mathcal{F}[f'](k) = \int_{-\infty}^{\infty} f'(x) e^{ikx} dx.$$

Integrate by parts with $u = e^{ikx}$ and $dv = f'(x) dx$, so $du = ik e^{ikx} dx$ and $v = f(x)$:

$$\int_{-\infty}^{\infty} f'(x) e^{ikx} dx = \left[f(x) e^{ikx} \right]_{-\infty}^{\infty} - ik \int_{-\infty}^{\infty} f(x) e^{ikx} dx.$$

Boundary term vanishes. Since $|e^{ikx}| = 1$ for real k and $\lim_{x \rightarrow \pm\infty} f(x) = 0$,

$$\left[f(x) e^{ikx} \right]_{-\infty}^{\infty} = 0.$$

Identify the remaining integral as $\hat{f}(k)$:

$$\mathcal{F}[f'](k) = -ik \int_{-\infty}^{\infty} f(x) e^{ikx} dx = -ik \hat{f}(k). \quad \square$$

Remark. Applying the property n times gives the general result $\mathcal{F}[f^{(n)}](k) = (-ik)^n \hat{f}(k)$. In particular, $\mathcal{F}[f''](k) = -k^2 \hat{f}(k)$.

Part 2: Application to a Specific Function

Choice of function

Let

$$f(x) = \begin{cases} e^{-ax}, & x \geq 0, \\ 0, & x < 0, \end{cases} \quad a > 0.$$

This function is absolutely integrable, decays to zero as $x \rightarrow +\infty$, and vanishes identically for $x < 0$.

Step 1: Compute $\hat{f}(k)$ directly

$$\hat{f}(k) = \int_0^\infty e^{-ax} e^{ikx} dx = \int_0^\infty e^{-(a-ik)x} dx.$$

Since $a > 0$ ensures convergence:

$$\hat{f}(k) = \left[\frac{e^{-(a-ik)x}}{-(a-ik)} \right]_0^\infty = 0 - \frac{1}{-(a-ik)} = \frac{1}{a-ik}.$$

Step 2: Find $\mathcal{F}[f'](k)$ using the derivative property

By the theorem:

$$\mathcal{F}[f'](k) = -ik \hat{f}(k) = \frac{-ik}{a-ik}.$$

Step 3: Verify by computing $\mathcal{F}[f'](k)$ directly

For $x > 0$, $f'(x) = -ae^{-ax}$. At $x = 0$, the function jumps from 0 to 1 (since $f(0^-) = 0$ and $f(0^+) = 1$), contributing a Dirac delta term. In the distributional sense:

$$f'(x) = -ae^{-ax} \mathbf{1}_{x \geq 0} + \delta(x).$$

Taking the Fourier transform:

$$\begin{aligned} \mathcal{F}[f'](k) &= \int_0^\infty (-a) e^{-ax} e^{ikx} dx + \int_{-\infty}^\infty \delta(x) e^{ikx} dx \\ &= -a \cdot \frac{1}{a-ik} + 1 \\ &= \frac{-a + (a-ik)}{a-ik} = \frac{-ik}{a-ik}. \end{aligned}$$

This matches the result from Step 2 exactly. ✓

Part 3: Numerical Verification

Set $a = 2$ and $k = 1$.

Step A: Compute $\hat{f}(1)$

$$\hat{f}(1) = \frac{1}{2-i} = \frac{2+i}{2^2+1^2} = \frac{2+i}{5} = 0.4 + 0.2i.$$

Step B: Apply derivative property

$$-ik \hat{f}(k) \Big|_{k=1} = -i(1)(0.4 + 0.2i) = -0.4i - 0.2i^2 = 0.2 - 0.4i.$$

Step C: Compute $\mathcal{F}[f'](1)$ directly

$$\frac{-ik}{a - ik} \Big|_{a=2, k=1} = \frac{-i}{2 - i} = \frac{-i(2 + i)}{5} = \frac{-2i + 1}{5} = 0.2 - 0.4i.$$

Result: Both methods give $\mathcal{F}[f'](1) = 0.2 - 0.4i$. ✓

Verify also at $k = 2$, $a = 2$:

$$\hat{f}(2) = \frac{1}{2 - 2i} = \frac{2 + 2i}{8} = 0.25 + 0.25i.$$

$$-ik \hat{f}(k) \Big|_{k=2} = -2i(0.25 + 0.25i) = -0.5i - 0.5i^2 = 0.5 - 0.5i.$$

$$\frac{-ik}{a - ik} \Big|_{a=2, k=2} = \frac{-2i}{2 - 2i} = \frac{-2i(2 + 2i)}{8} = \frac{4 - 4i}{8} = 0.5 - 0.5i. \quad \checkmark$$

Conclusion

The derivative property $\mathcal{F}[f'](k) = -ik \hat{f}(k)$ was:

1. **Derived analytically** via integration by parts, with the boundary term vanishing due to the decay condition on f .
2. **Applied** to $f(x) = e^{-ax} \mathbf{1}_{x \geq 0}$, giving $\mathcal{F}[f'](k) = -ik/(a - ik)$.
3. **Verified numerically** at $(a, k) = (2, 1)$ and $(2, 2)$, with both the direct and property-based methods agreeing exactly.